

# Cheng-Hsiang Chiu

<https://cheng-hsiang-chiu.github.io/>

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## QUALIFICATION

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- Ph.D. with expertise in parallel and heterogeneous computing and task-graph scheduling
- Programming skills in C++, CUDA, and parallel frameworks with a focus on high-performance system design
- Proven ability to collaborate with cross-disciplinary teams and translate research into efficient solutions

## EDUCATION

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- **University of Wisconsin-Madison** Madison, Wisconsin, USA  
*Ph.D. in Electrical and Computer Engineering* Aug. 2023 – Aug. 2025
- **École Polytechnique Fédérale de Lausanne** Lausanne, Switzerland  
*M.S. in Computer Science* Sep. 2013 – Feb. 2016
- **National Chiao Tung University** Hsinchu, Taiwan  
*M.S. in Communications Engineering* Sep. 2005 – Aug. 2007
- **National Chung Cheng University** Chiayi, Taiwan  
*B.S. in Electrical Engineering* Sep. 2001 – Jun. 2005

## RESEARCH

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- **RL Scheduling**: Developed RL scheduling algorithms for work-stealing runtime and CUDA Graph
- **Taskflow**: Developed Pipeflow, AsyncTask, syclflow in Taskflow (<https://taskflow.github.io>).

## SELECTED EXPERIENCE

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- **Software Intern at Intel** Austin, Texas, USA  
*Developed the implicit SYCL Graph programming model* May 2022 - Aug. 2022
- **Software Intern at Cadence** Austin, Texas, USA  
*Accelerated the executions of buffer insertion algorithm by 16%* May 2021 - Aug. 2021

## SELECTED PUBLICATIONS

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- **C.H. Chiu**, C. Morchdi, C.C. Chang, C. Yu, Y. Zhou, and T.W. Huang, "Optimizing CUDA Graph Scheduling with Reinforcement Learning - A Case Study in SSTA Propagation," *MLCAD*, 2025
- **C.H. Chiu**, C. Morchdi, Y. Zhou, B. Zhang, C. Chang, and T.W. Huang, "Reinforcement Learning-generated Topological Order for Dynamic Task Graph Scheduling", *HPEC*, 2024.
- C. Morchdi, **C.H. Chiu**, Y. Zhou, and T.W. Huang, "A Resource-efficient Task Scheduling System using Reinforcement Learning," *ASP-DAC*, 2024
- **C.H. Chiu**, Z. Xiong, Z. Guo, T.W. Huang, and Y. Lin, "An Efficient Task-parallel Pipeline Programming Framework," *HPC Asia*, 2024
- **C.H. Chiu**, D.L. Lin, and T.W. Huang, "Programming Dynamic Task Parallelism for Heterogeneous EDA Algorithms," *ICCAD*, 2023

## SKILLS

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- **Programming Language**: C, C++, Python, HTML, SQL
- **Programming Model**: Taskflow, SYCL, CUDA, OpenMP, MPI, PyTorch
- **Unit Test**: doctest
- **Profiler**: gprof, perf, Valgrind
- **DevOps**: GitHub, CI/CD, Docker
- **Build Tools**: CMake